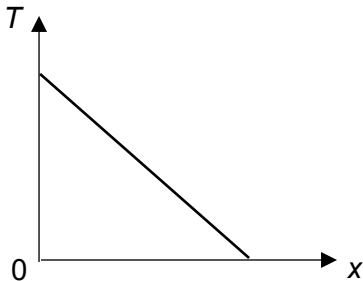
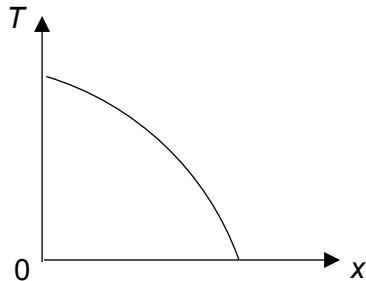
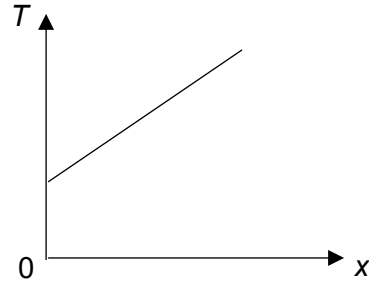


3

A body moves with a constant acceleration opposite in direction to its initial velocity. Which one of the following graphs represents the variation of its kinetic energy T with the distance travelled x from its initial position?

A**B****[Turn over** T ↑

	C	D



Ans: A

$v^2 = u^2 + 2as = u^2 - 2ax$ (negative sign to indicate direction of acceleration opposite of displacement)

$$\frac{1}{2}mv^2 = \frac{1}{2}mu^2 - max$$

So graph kinetic energy against x is straight line with negative gradient (also can be deduced using conservation of energy)